Chapter 11 Warm-Up: From Dictionaries to Classes

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1 Review: Dictionary Basics

A Python dictionary stores information as key–value pairs. For example,

```
students = {2601001: "Ana", 2601002: "Ben"}
```

Here, each student ID is a **key** and each name is its corresponding **value**. What is printed by `print(students[2601002])`? Add to the dictionary a student named Diego with ID 2601008. A dictionary is useful when we want to associate one piece of information with another. A **class** goes further: it lets us define a new kind of object that can store several pieces of information and also define what that object can **do**.

2 Complex Numbers

A complex number has a **real part** and an **imaginary part**:

$$z = a + bj \tag{1}$$

where

$$j^2 = -1 \tag{2}$$

To add complex numbers, add real parts together and imaginary parts together:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Subtraction works the same way:

$$(a + bi) - (c + di) = (a - c) + (b - d)i.$$

3 A Class for Complex Numbers

We can create our own Python class that stores the real and imaginary parts of a complex number.

```
class ComplexNumber:
    def __init__(self, real, imag):
        self.real = real
        self.imag = imag
```

Now we can create two **instances**:

```
z1 = ComplexNumber(3, 2)
z2 = ComplexNumber(5, -4)
```

For `z1`, the values are

```
z1.real  ->  3
z1.imag  ->  2
```

In your own words, what does `self` represent?

4 Teaching the Class to Add and Subtract

Normally Python does not know how two objects of a new class should be added. We can define that behavior with the special method `__add__()`:

```
class ComplexNumber:
    def __init__(self, real, imag):
        self.real = real
        self.imag = imag

    def __add__(self, other):
        return ComplexNumber(
            self.real + other.real,
            self.imag + other.imag
        )
```

Now Python understands

```
z3 = z1 + z2
```

because `__add__()` tells Python what the + operator means for two `ComplexNumber` objects. Now add a `__sub__()` method so that Python understands knows what the − means.

5 Check Your Understanding

1. Let $z_1 = 2 + 2j$, and $z_2 = 2 - 2j$. Using your class, compute (a) $z_1 + z_2$, and (b) $z_1 - z_2$.
2. Let $z_1 = 3 + 0j$, and $z_2 = 0 + 3j$. Using your class, compute (a) $z_1 + z_2$, and (b) $z_1 - z_2$.
3. How would you define a method within your `ComplexNumbers` class to handle *multiplication*?